

Phase 2 — self-consistent RPA, peak memory and wall time

Mercury (Z=80), all-electron, one outer cycle. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Self-consistent-potential RPA (xc='RPA'), corrected code vs original.

Radial mesh sweep (Lmax = 30)

N_fe	n_quad	new (MiB)	orig (MiB)	reduction	new (s)	orig (s)	speedup	dE (Ha)
10	450	263	-	-	17.8	-	-	-
15	675	375	-	-	45.7	-	-	-
25	1125	650	-	-	149.4	-	-	-
40	1800	1390	-	-	460.8	-	-	-

Angular sweep (N_fe = 40)

Lmax	n_quad	new (MiB)	orig (MiB)	reduction	new (s)	orig (s)	speedup	dE (Ha)
5	1800	1138	-	-	107.2	-	-	-
10	1800	1154	-	-	177.7	-	-	-
20	1800	1258	-	-	314.4	-	-	-
30	1800	1390	-	-	460.8	-	-	-

reduction = (new - orig)/orig on peak memory speedup = orig/new on wall time dE = new - orig on total energy
Peak memory is the sampled maximum resident set size; kernel ru_maxrss and SLURM MaxRSS agree with it to ~1%.